assignment1

October 27, 2026

0.1 Statistical Analysis

0.2 Assignment 1

0.3 PRATIK SUNIL ABHANG

0.4 260740150018

[45]:
```
import pandas as pd
```

[46]:
```
df = pd.read_csv(r"E:\GITHUB\CDAC-ACC-HPC-AI-2026\Statistical␣
↪Analysis\datasets\Housing.csv")
```

[47]:
```
df
```

[47]:

	price	lotsize	bedrooms	bathrms	stories	driveway	recroom	fullbase	gashw	airco	garagepl	prefarea
0	42000.0	5850	3	1	2	yes	no	yes	no	no	1	no
1	38500.0	4000	2	1	1	yes	no	no	no	no	0	no
2	49500.0	3060	3	1	1	yes	no	no	no	no	0	no
3	60500.0	6650	3	1	2	yes	yes	no	no	no	0	no
4	61000.0	6360	2	1	1	yes	no	no	no	no	0	no
..	...	...	...	...	...	...	...	...	...	...	...	...
541	91500.0	4800	3	2	4	yes	yes	no	no	yes	0	no
542	94000.0	6000	3	2	4	yes	no	no	no	yes	0	no
543	103000.0	6000	3	2	4	yes	yes	no	no	yes	1	no
544	105000.0	6000	3	2	2	yes	yes	no	no	yes	1	no
545	105000.0	6000	3	1	2	yes	no	no	no	yes	1	no

```
[546 rows x 12 columns]
```

0.5 1. Mean, Standard Deviation, Median and Quartiles

0.6 LOTSIZE

[48]:
```
print("Mean:", df["lotsize"].mean())
```

```
Mean: 5150.2655677655675
```

[49]:
```
print("Standard Deviation:", df["lotsize"].std())
```

```
Standard Deviation: 2168.158725228557
```

[50]:
```
print("Median:", df["lotsize"].median())
```

```
Median: 4600.0
```

[51]:
```
print("Q1:", df["lotsize"].quantile(0.25))
print("Q2:", df["lotsize"].quantile(0.50))
print("Q3:", df["lotsize"].quantile(0.75))
```

```
Q1: 3600.0
Q2: 4600.0
Q3: 6360.0
```

0.7 PRICE

[52]:
```
print("Mean:", df["price"].mean())
```

```
Mean: 68121.59706959708
```

[53]:
```
print("Standard Deviation:", df["price"].std())
```

```
Standard Deviation: 26702.670925794933
```

[54]:
```
print("Median:", df["price"].median())
```

```
Median: 62000.0
```

[55]:
```
print("Q1:", df["price"].quantile(0.25))
print("Q2:", df["price"].quantile(0.50))
print("Q3:", df["price"].quantile(0.75))
```

```
Q1: 49125.0
Q2: 62000.0
Q3: 82000.0
```

0.8 2. Mean, Standard Deviation, Median and Quartiles Grouped by airco

0.9 LOTSIZE

[56]:
```
for group, data in df.groupby("airco"):
    print("\nAirco:", group)
    print("Mean:", data["lotsize"].mean())
    print("Standard Deviation:", data["lotsize"].std())
    print("Median:", data["lotsize"].median())
    print("Q1:", data["lotsize"].quantile(0.25))
    print("Q2:", data["lotsize"].quantile(0.50))
    print("Q3:", data["lotsize"].quantile(0.75))
```

```
Airco: no
Mean: 4823.109919571046
Standard Deviation: 2178.794880285505
Median: 4050.0
Q1: 3450.0
Q2: 4050.0
Q3: 5985.0

Airco: yes
Mean: 5855.635838150289
Standard Deviation: 1973.7525999773788
Median: 6000.0
Q1: 4510.0
Q2: 6000.0
Q3: 6600.0
```

0.10 PRICE

[57]:
```
for group, data in df.groupby("airco"):
    print("\nAirco:", group)
    print("Mean:", data["price"].mean())
    print("Standard Deviation:", data["price"].std())
    print("Median:", data["price"].median())
    print("Q1:", data["price"].quantile(0.25))
    print("Q2:", data["price"].quantile(0.50))
    print("Q3:", data["price"].quantile(0.75))
```

```
Airco: no
Mean: 59884.85254691689
Standard Deviation: 21338.739442297516
Median: 55500.0
Q1: 45000.0
Q2: 55500.0
Q3: 70000.0
```

```
Airco: yes
Mean: 85880.58959537573
Standard Deviation: 28463.438868691737
Median: 82000.0
Q1: 66000.0
Q2: 82000.0
Q3: 103000.0
```

0.11 3: Mean, Standard Deviation, Median and Quartiles Grouped by prefarea

[58]:
```
for group, data in df.groupby("prefarea"):
    print("\nPrefarea:", group)
    print("Mean:", data["lotsize"].mean())
    print("Standard Deviation:", data["lotsize"].std())
    print("Median:", data["lotsize"].median())
    print("Q1:", data["lotsize"].quantile(0.25))
    print("Q2:", data["lotsize"].quantile(0.50))
    print("Q3:", data["lotsize"].quantile(0.75))
```

```
Prefarea: no
Mean: 4868.832535885167
Standard Deviation: 2021.2010214253357
Median: 4310.0
Q1: 3500.0
Q2: 4310.0
Q3: 6000.0

Prefarea: yes
Mean: 6069.3203125
Standard Deviation: 2376.429949744121
Median: 6360.0
Q1: 4380.0
Q2: 6360.0
Q3: 7103.75
```

[59]:
```
for group, data in df.groupby("prefarea"):
    print("\nPrefarea:", group)
    print("Mean:", data["price"].mean())
    print("Standard Deviation:", data["price"].std())
    print("Median:", data["price"].median())
    print("Q1:", data["price"].quantile(0.25))
    print("Q2:", data["price"].quantile(0.50))
    print("Q3:", data["price"].quantile(0.75))
```

```
Prefarea: no
```

```
Mean: 63263.48564593301
Standard Deviation: 24214.231519553276
Median: 58500.0
Q1: 46625.0
Q2: 58500.0
Q3: 73000.0

Prefarea: yes
Mean: 83986.3671875
Standard Deviation: 28342.92663161682
Median: 79750.0
Q1: 64975.0
Q2: 79750.0
Q3: 95000.0
```

[ ]: